

Textbook EIA Work Progress Logging

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1 Uploading data from OECD only - OECD salaries

Before this work I did three later files:

1. textbook EIA work progress
2. textbook EIA work progress 2
3. textbook EIA statcan salaries

textbook EIA statcan salaries ends with a good agreement between prediction and reality.

1.1 Work done in statcan salaries

In *GDP_comp.py* I upload OECD data and Statcan data and integrate them by

1. Converting Statcan from CAD to USD
2. Mapping the Statcan sectors (more detailed) to OECD sectors

GDP_comp.py is the same as *func_upload_data.py* where I upload OECD and statcan data into *income_multipliers*.

plot_T_vs_time.py plots T of the 6 ICT sectors as function of year. Sector G (wholesaling) has variability in T .

OECD_statcan_comp.py compares GDP and II and output between OECD and statcan.

In *read_statcan.py* I upload all data from statcan - not just GDP. I need the “compensation of employees” for the closed model. I updated *func_data_upload.py* to *func_data_upload2.py* to include all data of statcan.

In *output_multipliers_all_sectors.py* I plot the predicted output versus the real output. The simple model gives very exact results.

upload_data.py - this is uploading only GDP from statcan, and not a function. The early version of *func_upload_data2.py* *output_multipliers_all_sectors.py* where I calculate multipliers and plot predictions *OECD_statcan_comp.py* comparing GDP and II and output between OECD and statcan.

C26_multipliers.py adding HFCE to the final demand vector, I get a nice agreement between prediction and reality. This file also has all the multipliers.

1.2 OECD salaries - initial state

I start this directory with the files

1. *OECD_OECD_compare.py*
2. *multipliers_OECD_salaries.py*

multipliers_OECD_salaries.py has all the up to date calculations: multipliers and predictions.

the next step is to calculate the impacts and to print to excel files

1.3 Impact multipliers (using OECD salaries)

impacts.py prints out the impacts nicely. excel file *impact_multipliers.xlsx*

impacts.py has everything that *multipliers_OECD_salaries.py*, plus a printing to excel of the multipliers of the impacts.

print2xls2.py takes all the printouts from before and prints to a single excel file the II, T, L matrices as well as the multipliers and the impacts.

1.4 print2xls1 - print2xls3 - printing to xls

print2xls3.py: prints all the three data: II, T, L matrices as well as multipliers and impacts on three sheets in one excel file. *func_data_upload_OECD_salaries.py* is updated to read either TTL or DOM OECD tables.

What happened is that Tanveer did his work on a TTL OECD data, which means total = domestic plus imports. I needed to update the files: *print2xls.py* and *upload_data_OECD_salaries.py* to be able to read and write TTL data as well as DOM (domestic) data.

1.5 *compare_Tanveer_matrices.py*

compare_Tanveer_matrices.py - to compare my matrices with Tanveers.

Even when the files are located in different directories, always open vscode from *Textbook_EIA* to run the files from there.

I removed *func_safe_divide* matrix and vector to an imported function I should delete them from any print2xls and other files and add from *func_safe_divide* import *safe_divide*, *safe_divide_vector* to the top.

Functions I have taken out:

1. *func_safe_divide* import *safe_divide*, *safe_divide_vector*

Already did this in

1. *print2xls3.py*
2. *compare_Tanveer_matrices.py*

I've checked and the matrices T , T_c and L_{df} are the same as Tanveers (using TTL OECD table).

Statcan Code	Description	OECD code
A	Agriculture, forestry and fishing	-
A01	Crop and animal production, hunting and related service activities	A01_02
A02	Forestry and logging	A01_02
A03	Fishing and aquaculture	A03
B	Mining and quarrying	-
B05	Mining of coal and lignite	B05_06
B06	Extraction of crude petroleum and natural gas	B05_06
B07	Mining of metal ores	B07_08
B08	Other mining and quarrying	B07_08
B09	Mining support service activities	B09
C	Manufacturing	-
C10	Manufacture of food products	C10T12
C11	Manufacture of beverages	C10T12
C12	Manufacture of tobacco products	C10T12
C10T12	Manufacture of food products; beverages and tobacco products	C10T12
C13	Manufacture of textiles	C13_15
C14	Manufacture of wearing apparel	C13_15
C15	Manufacture of leather and related products	C13_15
C16	Manufacture of wood and of products of wood and cork, except furniture	C16_18
C17	Manufacture of paper and paper products	C16_18
C18	Printing and reproduction of recorded media	C16_18
C19	Manufacture of coke and refined petroleum products	C19
C20	Manufacture of chemicals and chemical products	C20_21
C21	Manufacture of basic pharmaceutical products and pharmaceutical preparations	C20_21
C22	Manufacture of rubber and plastics products	C22_23
C23	Manufacture of other non-metallic mineral products	C22_23
C24	Manufacture of basic metals	C24_25
C25	Manufacture of fabricated metal products, except machinery and equipment	C24_25
C26	Manufacture of computer, electronic and optical products	C26_27
C27	Manufacture of electrical equipment	C26_27
C28	Manufacture of machinery and equipment n.e.c.	C28
C29	Manufacture of motor vehicles, trailers and semi-trailers	C29_30
C30	Manufacture of other transport equipment	C29_30
C31	Manufacture of furniture	C31_33
C32	Other manufacturing	C31_33
C33	Repair and installation of machinery and equipment	C31_33
D	Electricity, gas, steam and air conditioning supply	D
E	Water supply; sewerage, waste management and remediation activities	E
E36	Water collection, treatment and supply	
E37T39	Sewerage, waste management, remediation activities	
E38	Waste collection, treatment and disposal activities; materials recovery	
F	Construction	4 F
F41	Construction of buildings	
F42	Civil engineering	
F43	Specialized construction activities	

Statcan Code	Description	OECD code
I	Accommodation and food service activities	I
I55	Accommodation	
I56	Food and beverage service activities	
J	Information and communication	?
J58	Publishing activities	
J59	Motion picture, video and television programme production, sound recording and music publishing activities	
J60	Programming and broadcasting activities	J58T60
J61	Telecommunications	J61
J62	Computer programming, consultancy and related activities	
J62.63	Computer programming, consultancy, and information service activities	J62.63
J63	Information service activities	
K	Financial and insurance activities	K
K64	Financial service activities, except insurance and pension funding	
K65	Insurance, reinsurance and pension funding, except compulsory social security	
K66	Activities auxiliary to financial service and insurance activities	
L	Real estate activities	L
L68B	Real estate activities excluding imputed rents	
M	Professional, scientific and technical activities	M
M69	Legal and accounting activities	
M69.70	Legal and accounting activities; activities of head offices; management consultancy activities	
M70	Activities of head offices; management consultancy activities	
M71	Architectural and engineering activities; technical testing and analysis	
M72	Scientific research and development	
M73	Advertising and market research	
M74	Other professional, scientific and technical activities	
M74.75	Other professional, scientific and technical activities; veterinary activities	
N	Administrative and support service activities	N
N77	Rental and leasing activities	
N78	Employment activities	
N79	Travel agency, tour operator, reservation service and related activities	
N80	Security and investigation activities	
N80T82	Security and investigation, service and landscape, office administrative and support activities	
N81	Services to buildings and landscape activities	
N82	Office administrative, office support and other business support activities	
O	Public administration and defence; compulsory social security	O
P	Education	P
Q	Human health and social work activities	Q
Q86	Human health activities	
Q87	Residential care activities ⁵	
Q87.88	Residential care activities and social work activities without accommodation	
Q88	Social work activities without accommodation	